Data Manipulation with Pandas

A PRESNTATION REPORT

Submitted in the partial fulfilment for the Course of

DSA0405 - Fundamentals of Data Science
to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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JANUARY 2026

Report on Data Manipulation with Pandas

Introduction

This report is based on the presentation "Data Manipulation with Pandas". Pandas is a Python library used for data manipulation, cleaning, transformation, analysis and visualization.

Why Pandas

Pandas provides structured data handling, high performance through NumPy integration, compatibility with other data science libraries, and is widely used in industry.

Getting Started

A DataFrame is the primary data structure in Pandas. It stores data in rows and columns similar to a spreadsheet.

Selecting and Filtering

Data can be selected by columns, filtered using conditions, and accessed using loc and iloc.

Cleaning Data

Missing values can be detected, filled, or removed. Duplicate records can also be identified and deleted.

Transforming Data

Functions such as melt(), pivot(), merge(), and groupby() reshape and summarize datasets.

Advanced Operations

apply(), map(), replace(), transform(), and pivot_table() simplify custom transformations and reporting.

Time Series Analysis

Pandas supports date conversion, resampling, rolling averages, shift, and difference operations.

Visualization

Built-in plotting supports line, bar, histogram, and scatter charts using Matplotlib.

Best Practices

Use df.info() and df.describe() to inspect datasets, write readable code, and continue learning advanced tools like Dask and Polars.

Conclusion

Pandas is an essential library for efficient data analysis and preprocessing in Python and is widely used in real-world data science projects.